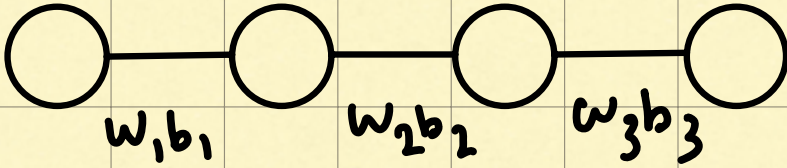


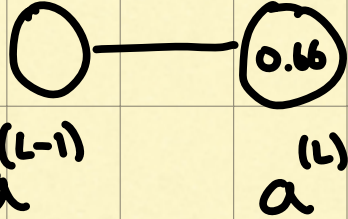
back propagation math

start w/ 1-1-1-1 neural network



how sensitive is $\text{cost}()$ to these variables?

So we know which adjustment will cause most efficient decrease in cost.

• focus on:  we desire 1.00.
 y

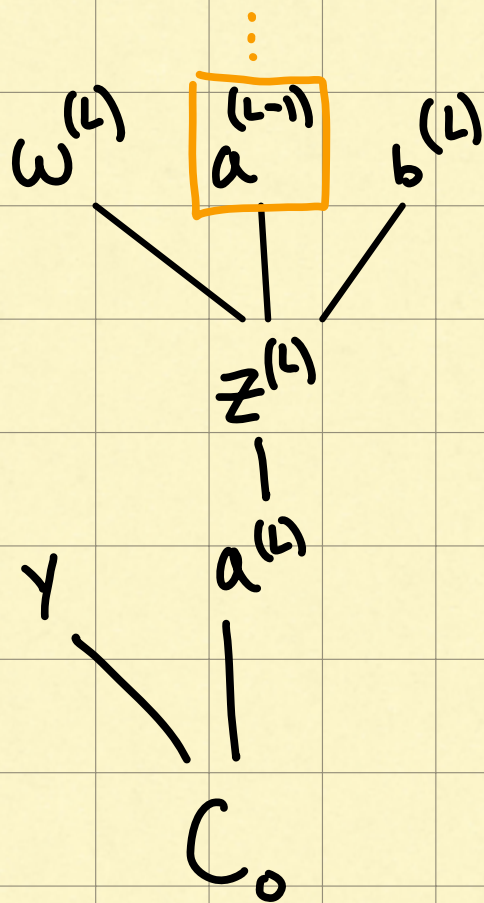
• Cost of simple network

$$C_0 = (a^{(L)} - y)^2$$

$$a^{(L)} = \sigma \left(w^{(L)} a^{(L-1)} + b^{(L)} \right)$$

$$a^{(L)} = \sigma \left(z^{(L)} \right)$$

relationships b/w terms.



Q: how sensitive is cost function to small changes in weight w_L ?

= derivative of C with respect to w_L ? $= \frac{\partial C_0}{\partial w^{(L)}}$

$$= \frac{\partial C_0}{\partial w^{(L)}} = \frac{\text{resulting nudge to cost}}{\text{tiny nudge to } w}$$

$\partial w^{(L)}$ causes $\partial z^{(L)}$ which causes nudge to $a^{(L)}$ which directly influences the cost

$$= \frac{\partial C_0}{\partial w^{(L)}} = \frac{\partial z^{(L)}}{\partial w^{(L)}} \cdot \frac{\partial a^{(L)}}{\partial z^{(L)}} \cdot \frac{\partial C_0}{\partial a^{(L)}} \quad \text{chain rule}$$

= C 's sensitivity to small changes to w .

Computing the derivatives:

$$\frac{\partial C_0}{\partial a^{(L)}} = 2(a^{(L)} - y) \quad // \text{even small changes to } a^{(L)} \text{ have big effects on the cost.}$$

$$\frac{\partial a^{(L)}}{\partial z^{(L)}} = \sigma'(z^{(L)}) //$$

$$\frac{\partial z^{(L)}}{\partial w^{(L)}} = a^{(L-1)} \quad // \begin{array}{l} \text{small nudge} \\ \text{to } w \end{array} \rightarrow \begin{array}{l} \text{affecting} \\ \text{last layer} \end{array} \quad \begin{array}{l} \text{depends on} \\ \text{last activation} \end{array}$$

$$\begin{aligned} \frac{\partial C_0}{\partial w^{(L)}} &= \frac{\partial z^{(L)}}{\partial w^{(L)}} \cdot \frac{\partial a^{(L)}}{\partial z^{(L)}} \cdot \frac{\partial C_0}{\partial a^{(L)}} \\ &= a^{(L-1)} \cdot \sigma'(z^{(L)}) \cdot 2(a^{(L)} - y) \end{aligned}$$

but all of this is how much 1 training example's cost (C_0) will change when changing $w^{(L)}$.

full cost function
w/ respect to w^L : = avg (cost of all training examples.)

$$\frac{\partial C}{\partial w^{(L)}} = \frac{1}{n} \sum_{k=0}^{n-1} \frac{\partial C_k}{\partial w^L}$$

⚡ full gradient is all weights & biases,
not just w^L :

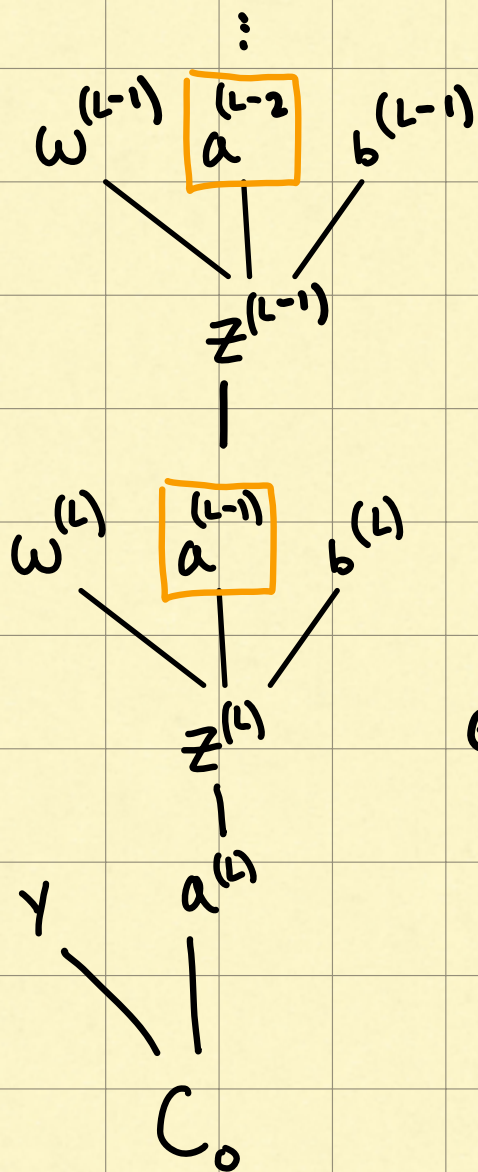
$$\nabla C = \begin{bmatrix} \frac{\partial C}{\partial w^{(1)}} \\ \frac{\partial C}{\partial b^{(1)}} \\ \vdots \\ \frac{\partial C}{\partial w^{(L)}} \\ \frac{\partial C}{\partial b^{(L)}} \end{bmatrix}$$

∇C
built up from
partial derivatives
of cost function
w/ respect to
all weights &
biases.

if you want: $\frac{\partial C_0}{\partial b^{(L)}} = \frac{\partial z^{(L)}}{\partial b^{(L)}} \cdot \dots = 1 \cdot \sigma'(z^{(L)}) \cdot 2(a^{(L)} - y)$

after little
nudge to b_L
instead

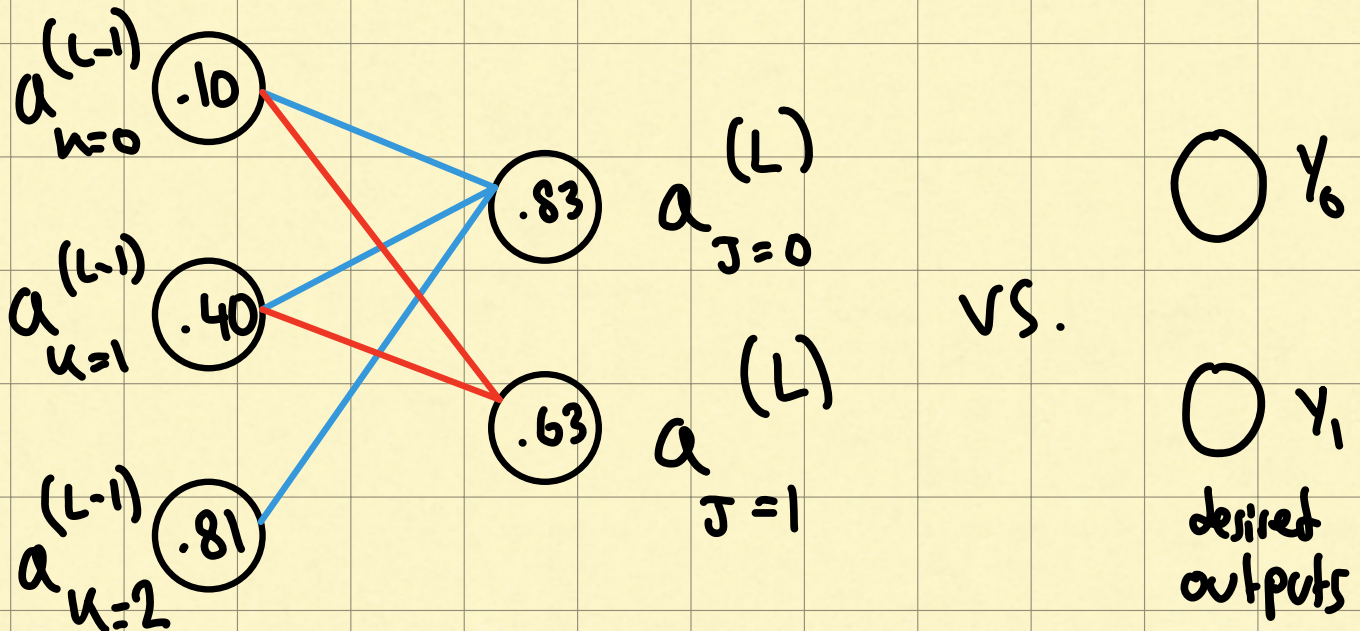
now for propagating backwards:



we can see how
sensitive cost function
is to activations in
previous layer from weights & biases.

ex:
$$\frac{\partial C_0}{\partial a^{(L-1)}} = \frac{\partial z^{(L)}}{\partial a^{(L-1)}} \cdot \frac{\partial a^{(L)}}{\partial z^{(L)}} \cdot \frac{\partial C_0}{\partial a^{(L)}}$$

what ab bigger network?



$$C_o = \sum_{j=0}^{n_{L-1}} (a_j^{(L)} - y_j)^2$$

add up squared diffs b/w
last layer & desired outputs

weight connected k-th neuron to J-th
neuron = $w_{jk}^{(L)}$

$$z_j^{(L)} = w_{j0}^{(L)} a_0^{(L-1)} + w_{j1}^{(L)} a_1^{(L-1)} + w_{j2}^{(L)} a_2^{(L-1)} + b_j^{(L)}$$

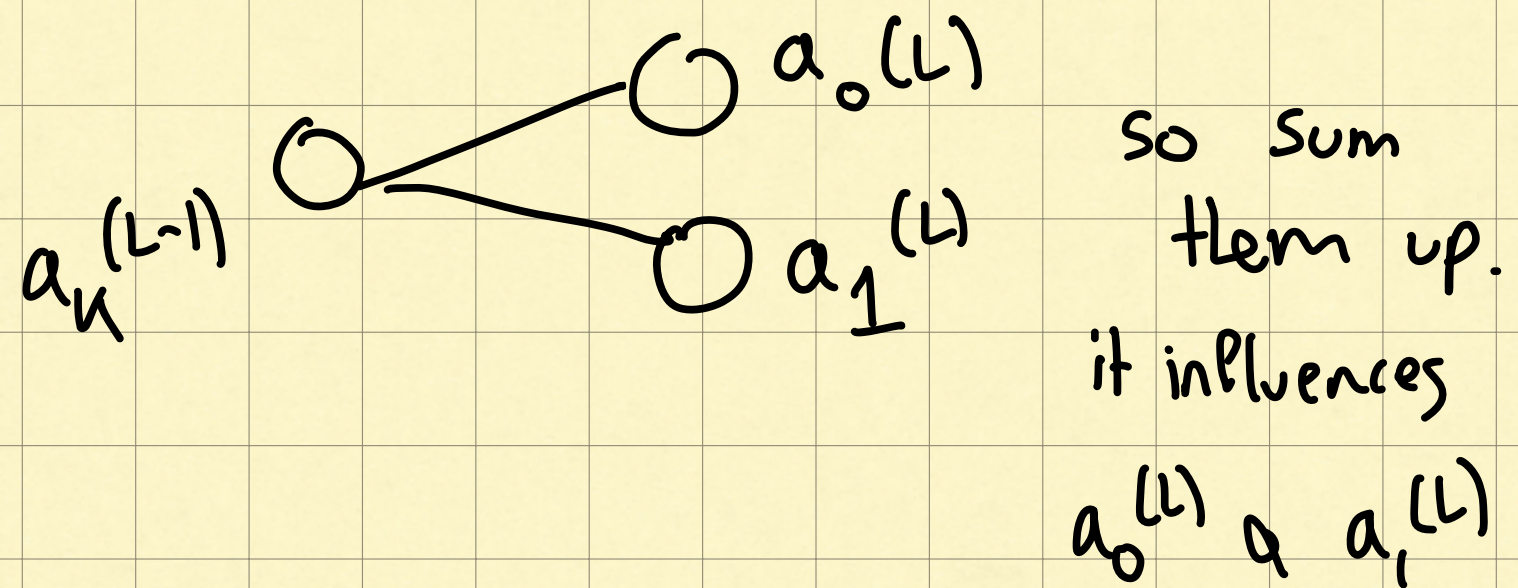
$$\text{so } a_j^{(L)} = \sigma(z_j^{(L)})$$

Same equations but more
subscripts

$$\frac{\partial C_0}{\partial w_{jk}^{(L)}} = \frac{\partial z_j^{(L)}}{\partial w_{jk}^{(L)}} \cdot \frac{\partial a_j^{(L)}}{\partial z_j^{(L)}} \cdot \frac{\partial C_0}{\partial a_j^{(L)}}$$

$\frac{\partial C_0}{\partial a_{jk}^{(L-1)}}$ = how much cost changes after little nudge to activations in previous layer

changes to $a_k^{(L-1)}$ affect cost function through multiple paths.



$$\frac{\partial C_0}{\partial a_{jk}^{(L-1)}} = \sum_{j=0}^{n_L-1} \frac{\partial z_j^{(L)}}{\partial a_k^{(L-1)}} \cdot \frac{\partial a_j^{(L)}}{\partial z_j^{(L)}} \cdot \frac{\partial C_0}{\partial a_j^{(L)}}$$

how sensitive cost function is
to activations in 2nd to last

layer, repeat for

all weights & biases

feeding into last
layer